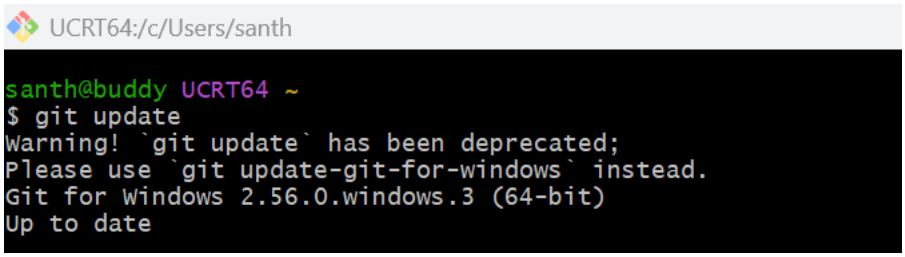
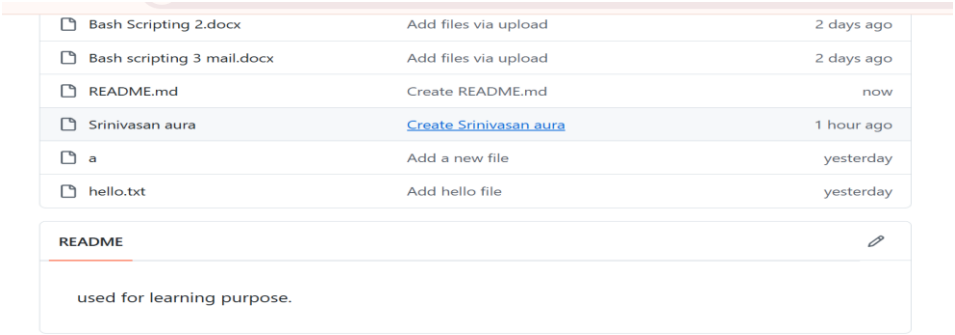
Git And Git Hub Task 2

1. Install git.



Download Git Run the installer Open Git Bash Verify Git installation and check for updates.

2. Create a repo in github with README.md and .ignore file.



- Create a new repository ,
- Check Add a README file.,
- Under Add .gitignore,
- Click Create repository

3. Clone the created repo to local.

```
santh@buddy UCRT64 ~
$ git clone git@github.com:srinivasanrajagopalan832-jpg/Srinivasan.git
fatal: destination path 'Srinivasan' already exists and is not an empty directory.

santh@buddy UCRT64 ~
$ git status
fatal: not a git repository (or any of the parent directories): .git

santh@buddy UCRT64 ~
$ cd ~/Srinivasan

santh@buddy UCRT64 ~/Srinivasan (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   b
        new file:   c
        new file:   d
        new file:   e
```

- Open Git Bash -Go to the location where you want to save your repository.-
- Click the green Code button and copy the ssh URl and paste in gitbash

4. Create two files in local repo.

```
    Add files via upload

santh@buddy UCRT64 ~/Srinivasan (main)
$ mkdir Sri

santh@buddy UCRT64 ~/Srinivasan (main)
$ touch sri.txt srini.txt

santh@buddy UCRT64 ~/Srinivasan (main)
$
```

- To create files in local repo we need to use the command touch.
- I have a file name sri.txt.

5. Commit two files and push to central Repository.

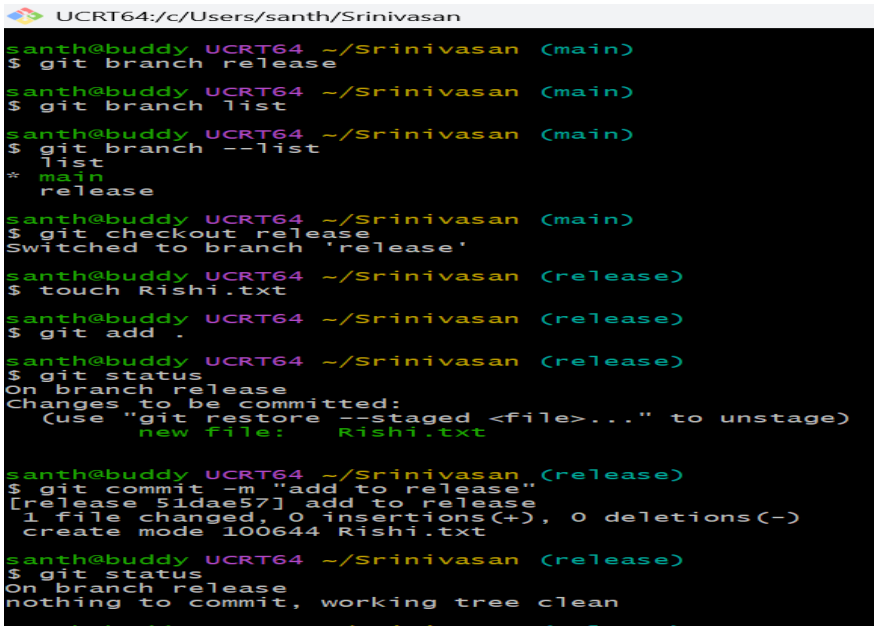
```
santh@buddy UCRT64 ~/Srinivasan (main)
$ git add .

santh@buddy UCRT64 ~/Srinivasan (main)
$ git commit -m "added to main branch"
[main 5f2caa5] added to main branch
 6 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 b
 create mode 100644 c
 create mode 100644 d
 create mode 100644 e
 create mode 100644 sri.txt
 create mode 100644 srini.txt
```

- Create two files in local repo and add a message in that file by using echo command

- And initialize git in that git config,
- Use git add . and git commit -m "Initial commit
- Use remote add orgin ssh – git branch -M main
- Need to pull origin and the git push to origin main

6. Create a branch in local and create a sample file and push to central.



- Create a new branch.
- -Create a sample file.
- -Add content to the file.
- Add and commit the file.
- Push the branch to GitHub

7. Create a branch in github and clone that to local.

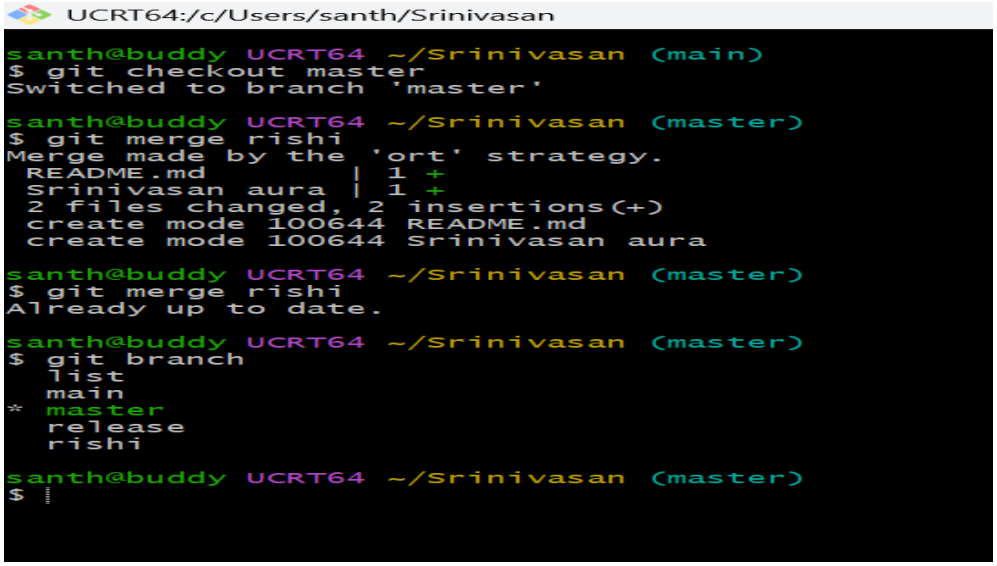
```
santh@buddy UCRT64 ~/Srinivasan (main)
$ git branch
  list
* main
  release

santh@buddy UCRT64 ~/Srinivasan (main)
$ git switch --track origin/rishi
branch 'rishi' set up to track 'origin/rishi'.
Switched to a new branch 'rishi'

santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git branch
  list
  main
  release
* rishi
```

- Create a branch in GitHub - Open Git Bash - Clone the repository
- Go into the repository - Get the latest branches
- Switch to the GitHub branch git clone
- downloads the repository to your computer. git fetch origin
- downloads information about remote branches.
- git checkout feature
- switches to the feature branch locally

8. Merge the created branch with master in git local.



• git branch Lists local branches

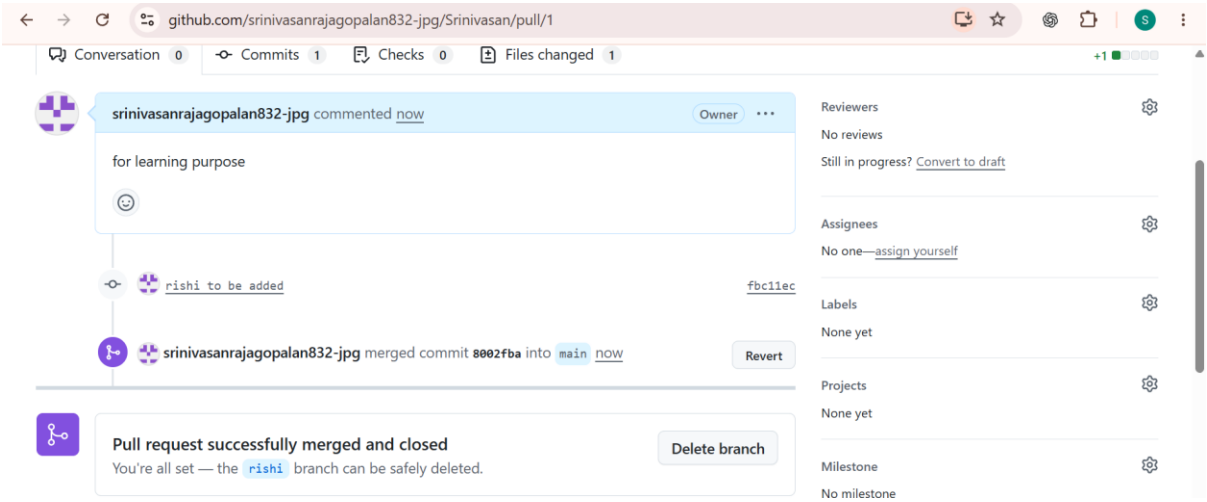
• git switch master Switches to master

• git merge release Merges the created branch into master

• git log -1 --graph --all Displays branch and commit history

- git push origin master Uploads the merged changes to GitHub

9. Merge the created branch with master in github by sending a pull request.



- Create a branch in your GitHub repository, feature -branch.
- Make changes to a file in the feature branch.
- Commit and push the changes to GitHub.
- Open your GitHub repository in the browser.
- Click the Pull requests tab.
- Click New pull request.
- Select the branches: Base: main — the branch that receives the changes.
- Compare: release — the branch containing your changes.
- Click Create pull request.
- Enter a title and description, then click Create pull request.
- Review the changes and click Merge pull request.
- Click Confirm merge to complete the process.

10. Create a file in local and send that to branch in github.

```
santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git checkout rishi
Already on 'rishi'
Your branch is up to date with 'origin/rishi'.

santh@buddy UCRT64 ~/Srinivasan (rishi)
$ echo "hello world" > vasan.txt

santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git add .
warning: in the working copy of 'vasan.txt', LF will be replaced by CRLF the next time Git touches it

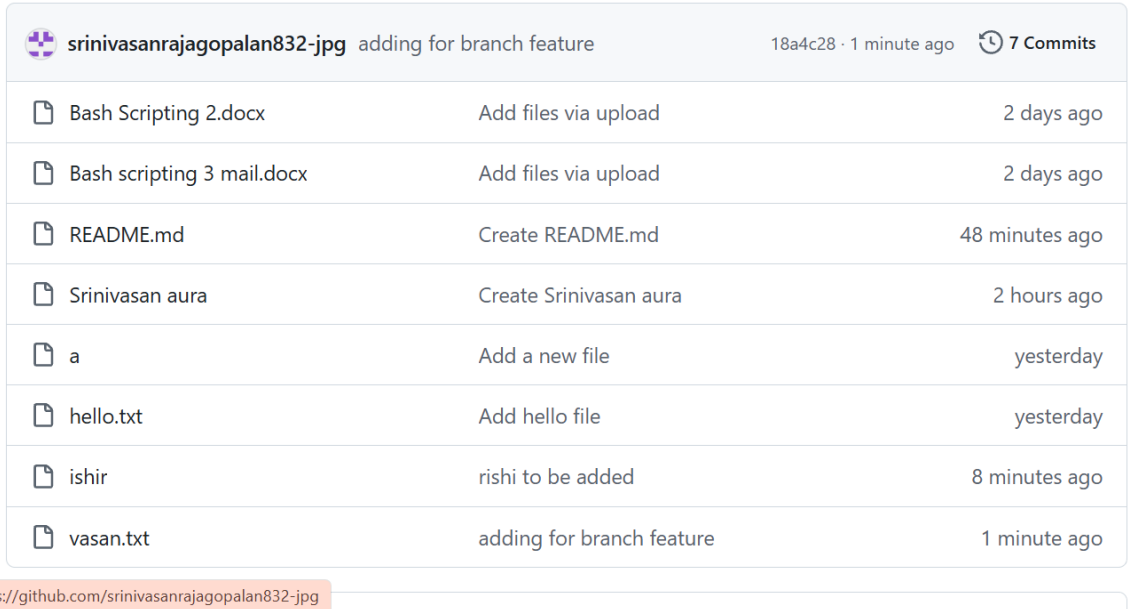
santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git status
On branch rishi
Your branch is up to date with 'origin/rishi'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   vasan.txt


santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git commit -m "adding for branch feature"
[rishi 18a4c28] adding for branch feature
 1 file changed, 1 insertion(+)
 create mode 100644 vasan.txt

santh@buddy UCRT64 ~/Srinivasan (rishi)
$ git push origin rishi
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 301.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:srinivasanrajagopalan832-jpg/Srinivasan.git
   fbc11ec..18a4c28  rishi -> rishi

santh@buddy UCRT64 ~/Srinivasan (rishi)
$
```



srinivasanrajagopalan832-jpg adding for branch feature — 18a4c28 · 1 minute ago — **7 Commits**

Bash Scripting 2.docx	Add files via upload	2 days ago
Bash scripting 3 mail.docx	Add files via upload	2 days ago
README.md	Create README.md	48 minutes ago
Srinivasan aura	Create Srinivasan aura	2 hours ago
a	Add a new file	yesterday
hello.txt	Add hello file	yesterday
ishir	rishi to be added	8 minutes ago
vasan.txt	adding for branch feature	1 minute ago

https://github.com/srinivasanrajagopalan832-jpg

• Switch to release branch • Create sample file • Read file cat sample.txt • Git add • Git commit –m • Git push –u origin release.

11.Clone only a branch from GitHub to local.

```
91966@VijayKumar MINGW64 ~/Desktop/tech (release)
$ git clone --branch stand --single-branch git@github.com:Gopichand15/tech.git
Cloning into 'tech'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 9 (delta 1), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (1/1), done.

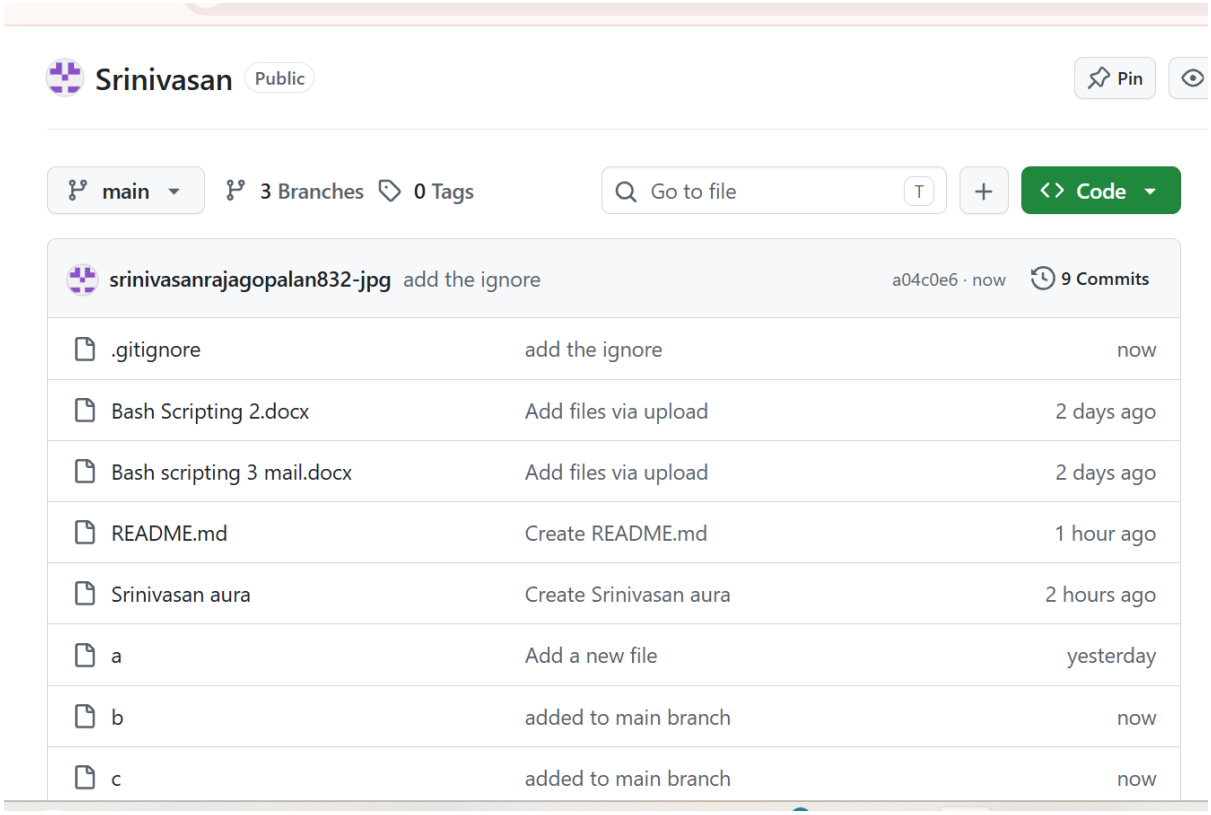
91966@VijayKumar MINGW64 ~/Desktop/tech (release)
$
```

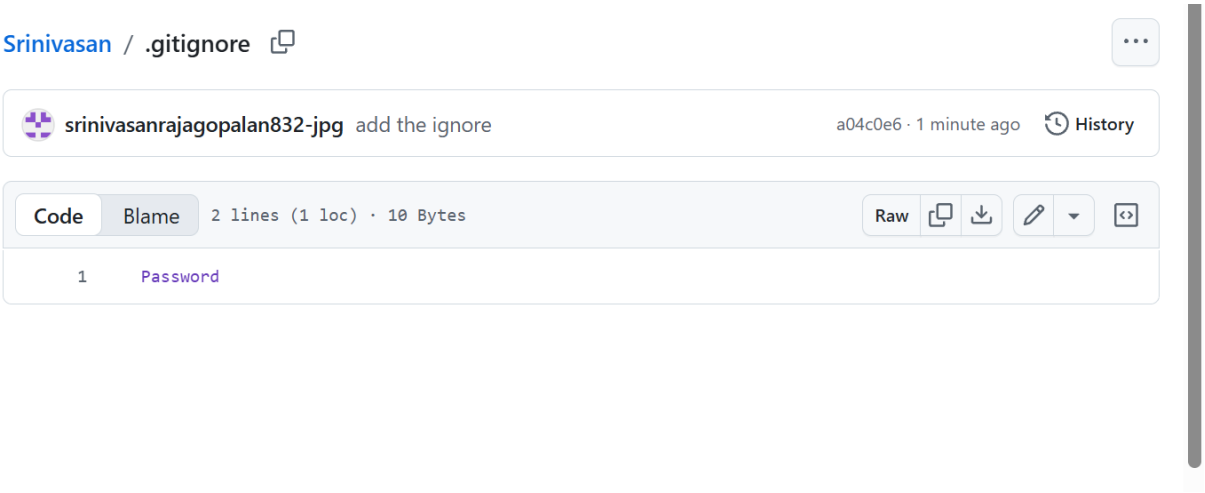
Clone a specific branch git clone --branch

feature-branch --single-branch REPOSITORY_URL

- Enter repository cd my-repository
- Check current branch git branch
- Check remote branches git branch -r

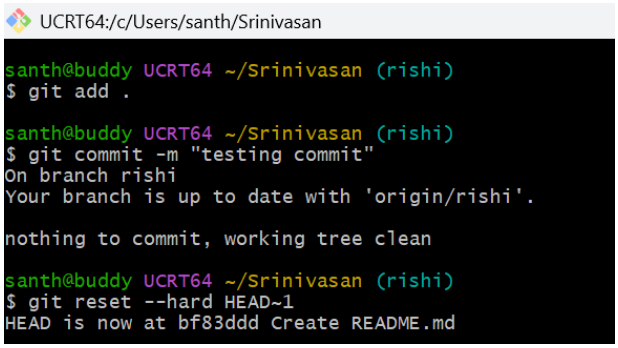
12.Create a file with all passwords and make that untraceable with git.





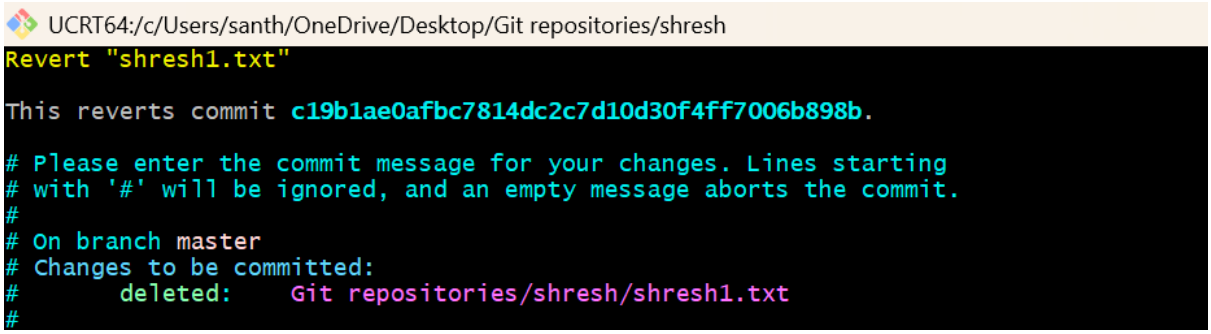
- Vi to edit the file. gitignore tells Git not to track matching files.
- git status checks whether Git is tracking the file.
- git commit saves the .gitignore change

13. Make a commit and make that commit reset without savings changes.



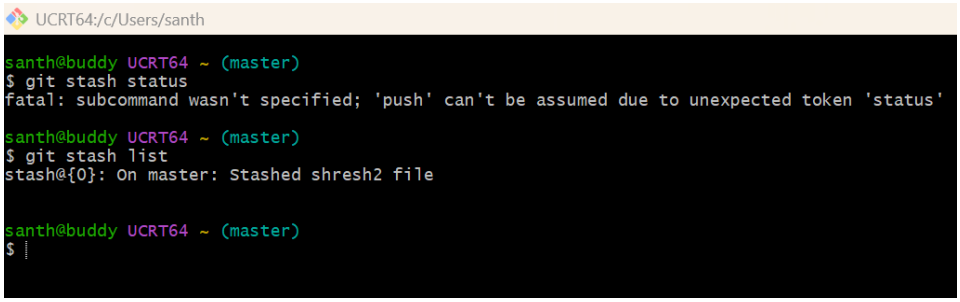
- echo "this is trayu file" > trayaksh – Creates a file with text.
- git add .
- Stages the file. git commit -m "Added trayu file "
- Saves the changes in a commit.
- git reset --hard HEAD~1 – Removes the latest local commit and discards its changes.
- git status – Checks the current repository status

14. Revert a commited commit to the older version.



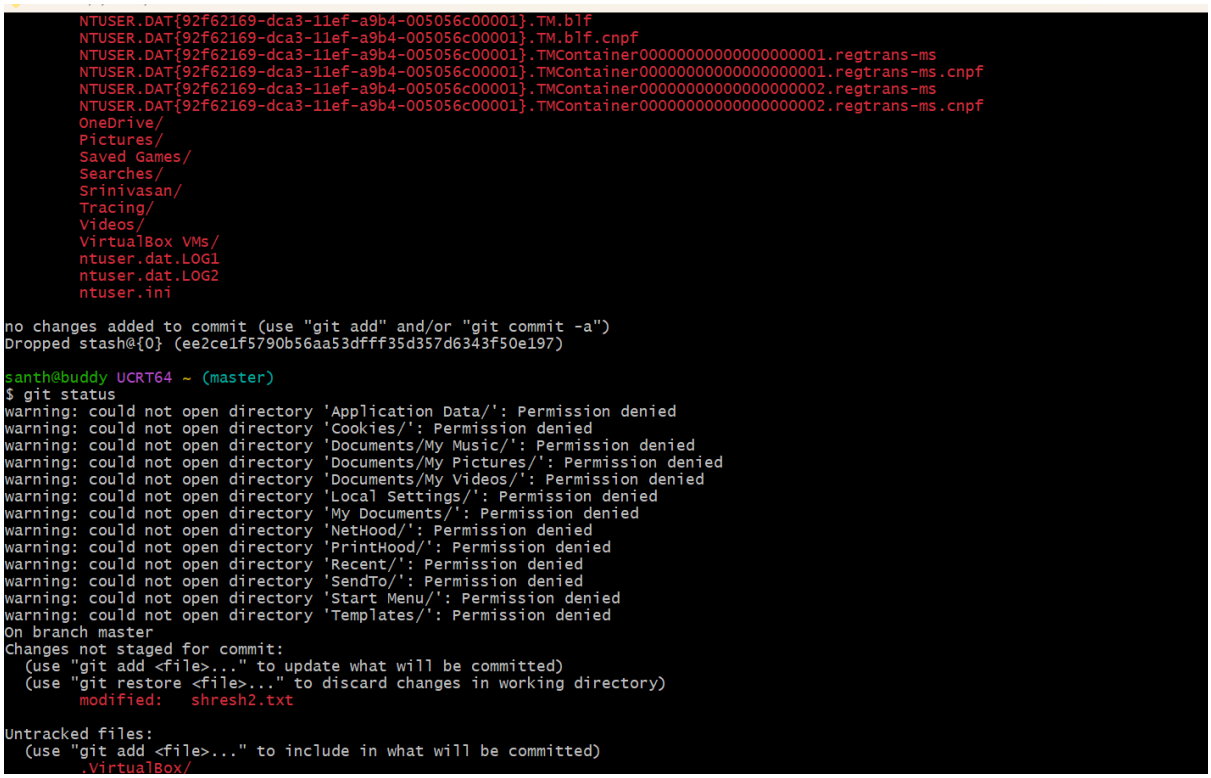
- git log --oneline – Displays the commit history.
- git revert HEAD – Undoes the latest commit by creating a new commit.
- git status – Checks the repository status.
- git log --oneline – Verifies the new revert commit

15. Push a file to stash without savings the changes and work on another file.



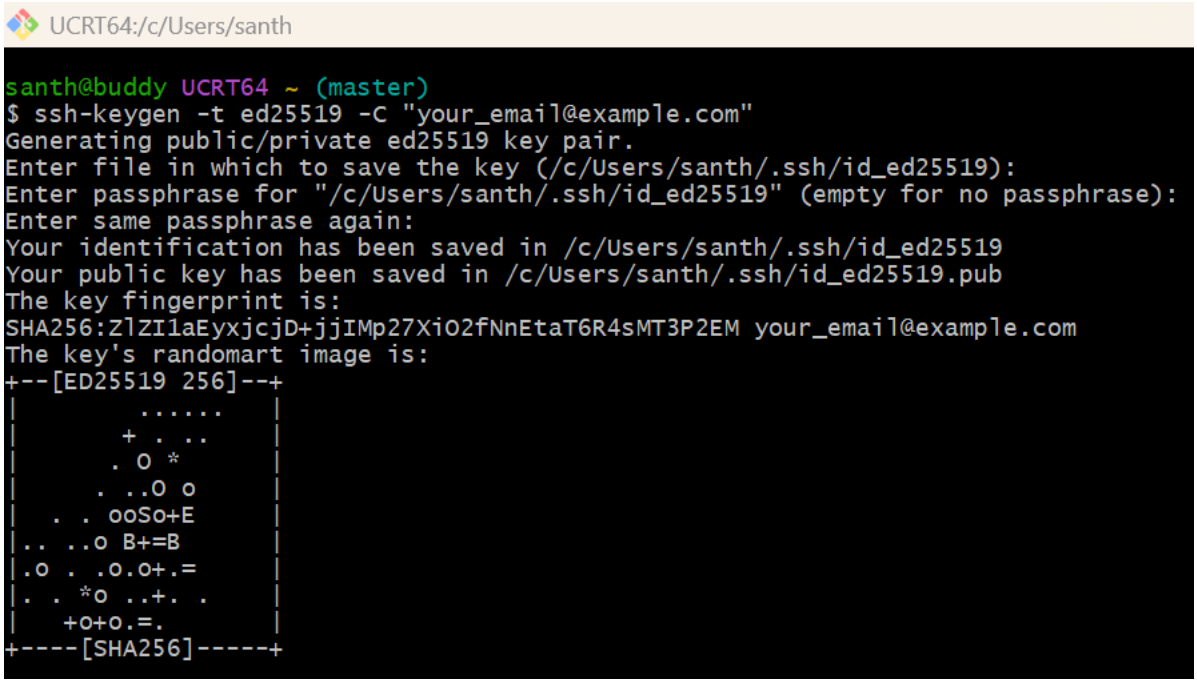
- git stash – Temporarily saves changes without committing them.
- git status – Checks the repository status.
- git stash pop – Restores the stashed changes.

16. Undo the stash file and start working on that again.



- Undo the stash file and start working on that again
- git stash pop is used to restore the changes saved in the stash and remove them from the stash list

17. Generate a ssh-keygen and configure into github.



• Generate key: Create your SSH key pair.

• Start agent: Start the SSH agent.

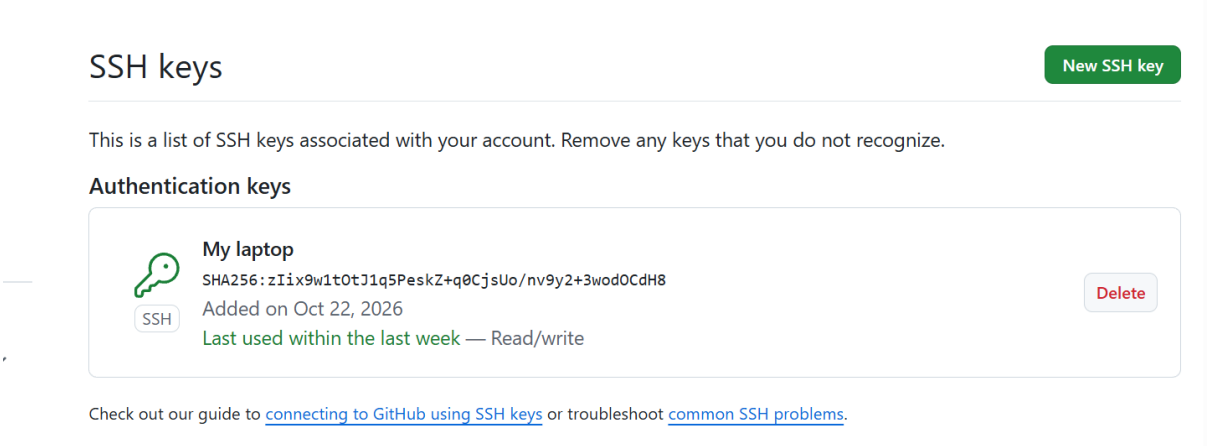
• Add key: Load your private key into the agent.

• Display public key: Copy the public key and add it under GitHub → Settings → SSH and GPG keys.

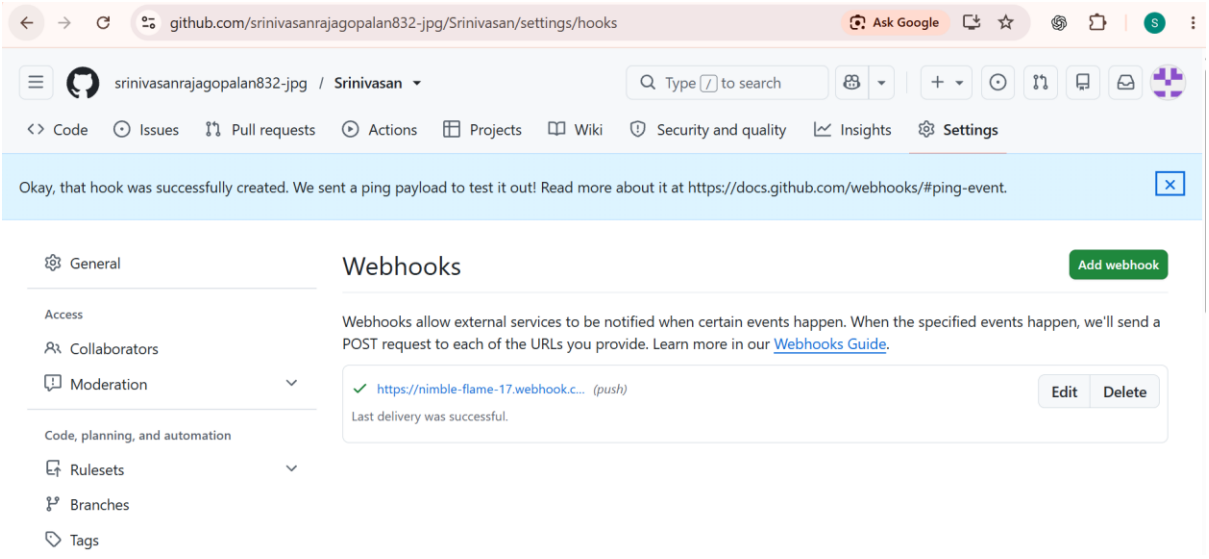
• Test connection: Run ssh -T git@github.com.

• Check remote: Run git remote -v to check your repository URL.

• Fetch changes: Run git fetch origin to retrieve remote updates.



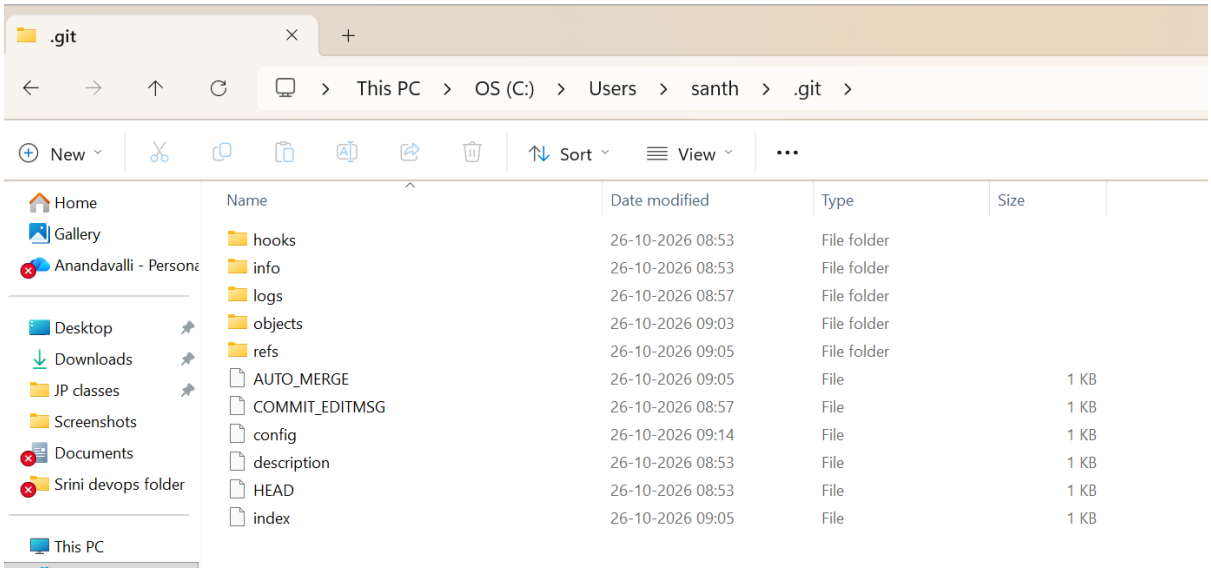
18. Configure webhooks to github.



- Configure webhooks to github.
- Open your GitHub repository.
- Go to Settings → Webhooks → Add webhook.
- Enter the Jenkins webhook URL in the Payload URL field.
- Select application/json. Choose Just the push event.

- Click Add webhook.

19. Basic understanding of .git file.



- The .git directory is the core of a local Git repository.
- If you delete it, your project files remain, but the repository's Git history and metadata are lost.
- Displays Git's internal files and folders.
- Important contents File/Directory Purpose HEAD Identifies the current branch or commit reference.
- config Stores repository-specific Git settings.
- objects/ Stores Git objects, including file contents and commits.
- refs/ Stores references to branches and tags.
- hooks/ Contains sample or configured scripts that run during Git events.
- index Stores the staging area's information.

20. Check all the logs of git.

```
santh@buddy UCRT64 ~ (master)
$ git log --stat
commit a4acc50b765ec114c01c3e9769c802c44724ab72 (HEAD -> master)
Author: Srinivasan <srinivasanrajagopalan832@gmail.com>
Date:   Mon Oct 26 08:57:07 2026 +0530

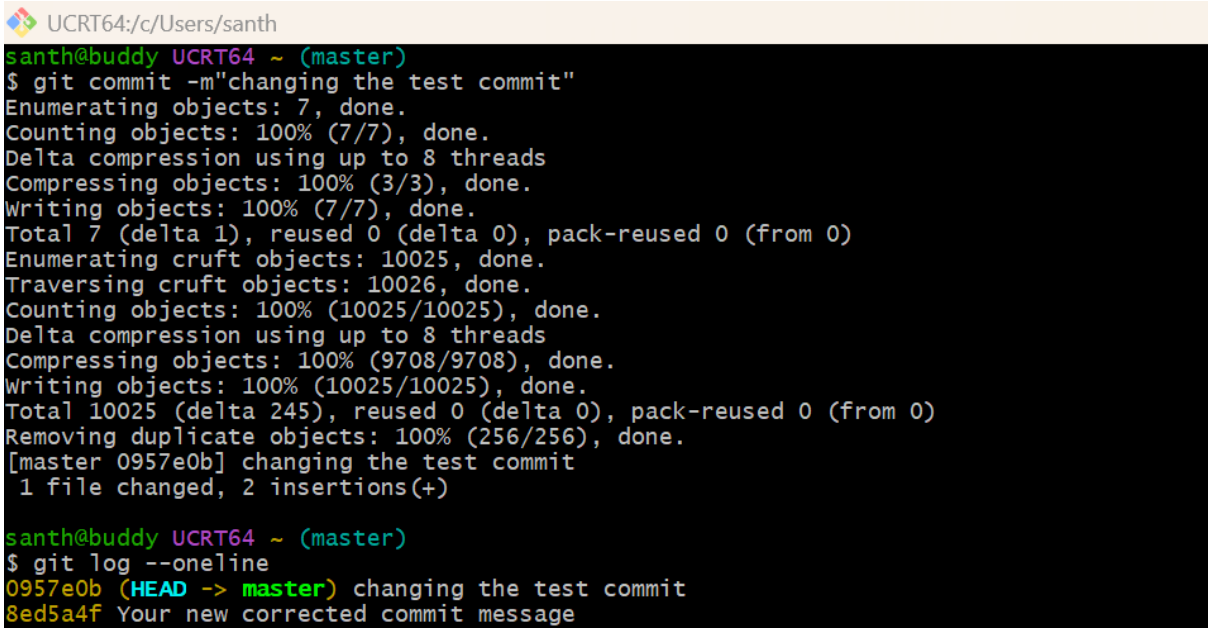
    shresh2.txt

 shresh2.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
```

- Use git log to check the commit history of your local repository.
- Git log Commit ID Author name Date and time Commit message

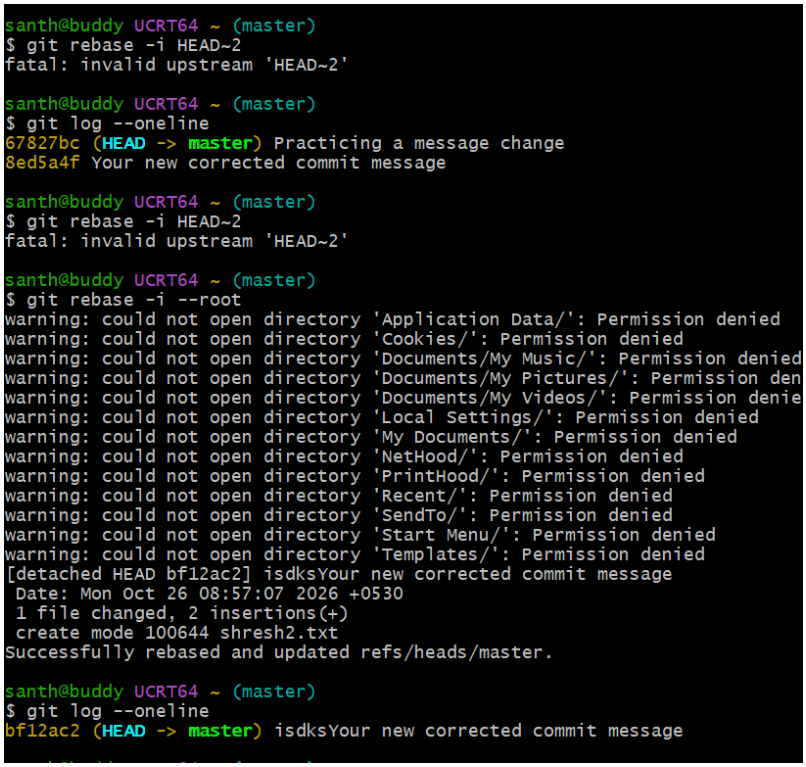
- git log --oneline --all –graph Shows commit history and branch merges in a graphical format.
- git log –all Displays commit history from all locally known branches.
- git log –oneline Shows a short summary of each commit.

21. Rename the commit message.



- git commit --amend -m is used to modify the latest commit message.
- Note: If the commit was already pushed to GitHub, updating it changes its commit ID.
- You may need to push with git push --force-with-lease; use caution if others share the branch.

22. Merge multiple commits into single commit.



- Check commit history - Start interactive rebase- Enter one message for the combined commit,
- then save and close the editor git rebase -i
- let you combine multiple commits into a single commit using squash.
- Note: This rewrites commit history.
- If those commits were already pushed to GitHub,
- coordinate with collaborators before pushing the rewritten history.